MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

DEPARTMENT OF CIVIL ENGINEERING			
END SEMESTER EXAMINATION		**NOV, 2024**	
Level	**B.Tech**	**Curriculum**	**2022**
Program Name	**B.Tech in Civil Engineering**	**Course Type**	**PE**
Course Name	**COASTAL ENGINEERING**	**Course Code**	**CIE 4076**
Semester	**VII**	**Max Marks**	**50**
Date	**22/11/2024**	**Duration**	**180 mins**

Instructions:
- **Answer ALL the questions.**
- **Data not provided may be suitably assumed**

Q. No.	Question	Marks	NBA		BL
			CO	PO	
1A	Calculate and Plot the distribution of maximum horizontal vertical water particle velocities across depth at 3 m intervals. The ocean wave measured at 9 m water depth has a wave height of 3 m and a wave period of 6 sec.	**3**	1	1,2,3	**4**
1B	Illustrate the surf similarity parameter and distinguish wave breaking based on the surf similarity parameter.	**2**	2	1,2	**3**
1C	Obtain dispersion relation using equation of velocity potential.	**5**	1	1,2,3	**3**
2A	Explain the available non-structural measures for coastal protection	**3**	3	1,2	**3**
2B	Infer if Sainflou's method can be adopted for a vertical wall breakwater design of 16 m height. The water depth at the site is 9 m with a wave period of 7 s. The refraction diagram gave spacing between adjacent orthogonals in deep water to be 2 units and at structure	**3**	3	1,2,3	**4**

<table>
<tr><td></td><td>site as 3 units. Consider shoaling coefficient of 0.9981 for a deepwater wave height of 4 m.</td><td></td><td></td><td></td><td></td></tr>
<tr><td>2C</td><td>Explain the difference between a port and a harbor. How are harbors classified? Explain with an example.</td><td>4</td><td>4</td><td>1,2</td><td>3</td></tr>
<tr><td>3A</td><td>What are slipways? Illustrate their role in vessel maintenance.</td><td>2</td><td>4</td><td>1,2,3</td><td>3</td></tr>
<tr><td>3B</td><td>Write a short note on :-
i) Floating dry dock ii) Dredging</td><td>4</td><td>4</td><td>1,2</td><td>3</td></tr>
<tr><td>3C</td><td>Interpret the importance of environment impact assessment during port construction.</td><td>4</td><td>4</td><td>1,2,3,6,7</td><td>3</td></tr>
<tr><td>4A</td><td>Infer how does coastal ecosystem gets affected by coastal pollution.</td><td>3</td><td>5</td><td>1,2,7</td><td>3</td></tr>
<tr><td>4B</td><td>Evaluate H_{mean}, $H_{1/10}$, $H_{1/3}$ and H_{rms} for the short term wave data recorded over 30 mins duration which yielded 125 waves.
<table>
<tr><td>H in m</td><td>0-0.5</td><td>0.6-1.1</td><td>1.2-1.7</td><td>1.8-2.3</td><td>2.4-2.9</td><td>3-3.5</td></tr>
<tr><td>No. of occurrences</td><td>25</td><td>33</td><td>28</td><td>22</td><td>14</td><td>3</td></tr>
</table></td><td>4</td><td>5</td><td>1,2,3</td><td>4</td></tr>
<tr><td>4C</td><td>With the help of neat diagram explain various types of berthing structures.</td><td>3</td><td>5</td><td>1,2</td><td>3</td></tr>
<tr><td>5A</td><td>Compare artificial armour blocks with Natural rocks and indicate the key differences.</td><td>3</td><td>5</td><td>1,2,3</td><td>2</td></tr>
<tr><td>5B</td><td>Design and draw a cross-section of truck of rubble mound breakwater at a location with following details
Mean water depth – 8 m
Design wave height – 3 m, Wave period-7 Sec
Wave runup - 3 m
Tidal correction – 1 m
Armour unit – Quarried natural rocks up to 100 kN size with Unit weight of 2.6 t/m3 can be used.
Take K_D – 3.2 and K_Δ - 1.</td><td>5</td><td>5</td><td>1,2,3</td><td>5</td></tr>
</table>

5C	Interpret the difference between Coastal Surveying and land surveying, mention the challenges	**2**	5	1,2,3	**3**